

Members: Alvin Dong, Dan Murlaga, John Luke Shulenberger

Party Members: John Luke

- Created Ally Class:
 - Act as entries into the larger party class
 - Modified data to include ally information including name, health, and damage
 - Allies include Satyr, Wizard, Dwarf, and Elf
 - Added allies to the location data, similar to how enemies are assigned to locations
 - Behaves in a similar way to the enemy class but attacks the enemy rather than the hero
- Created Party Class:
 - Allows the user to add allies that participate in combat alongside them
 - Players can add and remove allies from their party
 - Party is limited to two members to insure that the combat remains playable and does not become too overpowered.
 - Party members will then assist in combat and act as secondary attackers to the enemy, but battle still centers around whether the hero is alive.
 - Members can easily be changed to acquire a member with better attack or speed
- Added “talk” command:
 - How the classes are implemented into the main game loop
 - After each action player has an option to talk to potential allies
 - Will then print out a list of their party to remind them who is in their party
 - The player can choose to add that location’s ally to their party
 - If the player has reached their party limit, will prompt them to keep their same party or to replace one of their current party members.
- Modify the Combat loop:
 - Changed combat to allow for multiple combatants
 - Use a for loop to assign ally information to variables in the combat loop
 - Implement a conditional statement to determine which version of the combat loop to run depending on the number of allies in your party.
 - Functionality of the combat is largely the same, but when it is the ally’s turn to move they will automatically attack the enemy and the game will print out the damage done.

XP and Skills: Dan

- Implemented the XP mechanic
 - XP is gained through defeating enemies (equivalent to the enemy’s HP state)
 - Levels are gained through reaching XP milestones
 - Milestones start at 100 XP and increase by 1.5x (compoundly) with every level up

- Leveling up assigns 3 random stat growths between Max HP, Attack, Speed, and Luck
- Revamped/Introduced new stats
 - HP is separate from Max HP and allows the use of healing
 - Strength/Defense are in-battle stats that multiply dealt/received damage respectively
 - Luck is a multiplier that affects XP and Mana drops by killed enemies
 - Mana reworked to be a currency for using/learning skills (equivalent to the enemy's attack stat)
- Reworked the skill "tree"
 - Allows skills to have multiple pre-requisite skills, a level requirement, and base mana cost to learn
 - Divides skills into subtypes
 - Attack skills can be used in combat to replace basic attacks and have more base attack power but cost mana
 - Buff skills provide temporary boosts to Strength/Defense that last for one battle and have a cap of 1.75x
 - Heal skills can be used in and out of combat and allows recovery of current HP up to the limit of the Max HP stat
 - Passive skills cannot be used in combat and instead provide permanent stat growth
- Added UI elements
 - The help command now shows all current hero stats (HP, Max HP, Attack, Speed, Luck, Mana, XP)
 - The skill command (without any skill name) shows the list of all learned skills in addition to the list of all available skills (with level and mana requirements) based on the hero's prerequisites
- Added the ability to choose the hero's action during combat
 - Three actions are available to choose from:
 - (Basic) attack functions like the original combat system did, multiplying the hero's current attack stat by any Strength modifier
 - Skill (without any skill name) shows the list of all learned active skills, what their use costs are, and their effects
 - Skill (+ skill name) checks whether a certain skill can be used (mana cost, number of buff stacks, if current HP is less than Max HP) and uses its effect
 - If this is an attack skill, the base power is multiplied by any Strength modifier
 - Defend instantly ends the hero's turn while halving any received damage for one turn

Enemy AI + DFS: Alvin

- Enhanced enemy AI
 - Enemies will now make smarter decisions based on the current state of combat
 - For instance, some enemies might become more aggressive as your HP becomes lower
 - To adjust for this, some enemy stats have been tuned down
 - Enemies are also able to heal, block, or use elemental attacks
 - Enemies can target you, your allies, or everyone for attacks
- Quality of Life
 - You can now get a path from one point to another while outside of combat
 - Running “path [location 1] [location 2]” will generate a list of directions to follow that guarantees a valid path from the specified starting point to the specified ending point
 - You can omit the first location to have it find a path from the hero’s current location to the specified location
 - Note: This path is not always the shortest path.

ADTs:

Party ADT:

- Party() creates a new empty party, does not need any parameters
- add(member) adds a new member to the party, input is the member, returns the results of the operation (whether it was successful or if the limit has been reached)
- remove(member) removes the specified member from the party, input is the member
- replace(member, member) removes a member in the party and replaces it with a new member, requires both members
- check_limit() checks if the party has reached its member limit, needs no input, and returns a boolean value
- get_members() needs no parameters and returns a list of party members and their information
- print() prints the members of the party, needs no parameters

XP Bar:

- XPBar() creates an instance of an XP bar, and doesn't take parameters. Establishes a current XP and XP milestone stat which tracks progression.
- gain_xp() takes the amount of xp gained through combat and the hero that receives it and adds to their progress. If the progress reaches whatever the XP milestone is at the time, it calls the level_up() function. If the current level is 6, this function does nothing.
- level_up() only takes the hero that is leveling up as a parameter and increases the internal level and hero’s level by one while subtracting the milestone XP requirement from the current XP count. Every time this function is called, the XP milestone needed to level up again is multiplied by 1.5, and 3 random stats from Max HP, Attack, Speed, and

Luck are chosen and slightly incremented and fed back to the hero. If this function causes the level to increase to 6, a flag is triggered to prevent any more increases to XP.

- `__repr__()` gives a read out of the hero's current level and XP required to reach the next level or if the hero has reached max level.

Enemy Decision Tree:

- `getEnemyAction()` traverses the tree based on the nodes that it contains until it hits a leaf.